ONCOSENTINEL - SOFTWARE REQUIREMENTS SPECIFICATION

SRS v2.0 - Research-gap aligned

Version 2.0 | Research-gap aligned baseline | Body text 10 pt

Document Control

Field	Value
Document version	2.0
Status	Research-gap aligned baseline
Base body text	10 pt
Product	OncoSentinel
Primary scope	Multimodal cancer AI + clinical decision support + research evaluation
Taxonomy	Authoritative eight-category Multi Cancer taxonomy
Safety position	AI-assisted decision support; not autonomous diagnosis or treatment authority

1. Introduction

1.1 Purpose

This SRS defines the functional, non-functional, AI/ML, data, security, RAG, explainability, uncertainty, longitudinal modeling, fairness, clinical evaluation, integration, deployment, and governance requirements for OncoSentinel v2.0.

1.2 Research basis

The update is grounded in the supplied literature. Kourou et al. identify small datasets, inadequate validation, heterogeneous data, and limited clinical translation. Pati et al. demonstrate the value of federated multi-site training and explicit out-of-sample testing. Zhang et al. emphasize data normalization, interoperability, scale, and translation barriers. Sangwan demonstrates the need for quantitative explanation evaluation while identifying the limitations of pixel-level explanations. Shyr et al. demonstrate end-to-end rare-disease workflow modeling and identify the need for prospective evaluation, broader representation, and clinically meaningful deployment.

1.3 Scope

OncoSentinel shall accept text, images, PDFs and multimodal combinations; validate and normalize inputs; route supported cancer-analysis requests to appropriate ML models; route evidence questions to RAG; use a local LLM only under explicit grounding controls; support approved external fallback; provide uncertainty and safe abstention; create concept-based explanations; maintain longitudinal case timelines; support federated research; evaluate fairness; collect clinical outcomes; and preserve provenance/auditability.

1.4 Intended Users

- CLINICIAN
- RESEARCHER
- ADMIN
- ANALYST
- AUDITOR

- AUTHORIZED DATA CURATOR
- CLINICAL STUDY REVIEWER

2. Product Scope and Taxonomy

The initial ML scope remains the authoritative eight categories: Breast, Kidney, ALL/Leukemia, Oral Cancer, Lung & Colon, Lymphoma, Brain, and Cervical. Category-specific class mappings, modalities, datasets, calibration policies, uncertainty policies, explanation types, fairness strata, and clinical study status shall be versioned through registry/configuration.

3. System Architecture

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USER -> INPUT GATEWAY -> CONTENT UNDERSTANDING -> INTENT/RELEVANCE/SAFETY ROUTER
     -> ML ORCHESTRATOR -> SHARED BACKBONE -> CATEGORY HEAD -> UNCERTAINTY/OOD
     -> LONGITUDINAL CONTEXT -> CLINICAL XAI -> SAFETY/EVIDENCE/Fairness gates
     -> HUMAN REVIEW -> REPORT / OUTCOME CAPTURE
OR
     -> RAG ORCHESTRATOR -> VECTOR DB/KG -> EVIDENCE CHECK -> LOCAL LLM
     -> XAI/PROVENANCE/SAFETY -> HUMAN REVIEW -> REPORT / OUTCOME CAPTURE
OR
     -> APPROVED EXTERNAL WEB/API -> SOURCE VALIDATION -> LOCAL LLM -> VERIFIER
ALL ROUTES -> AUDIT + MODEL/DATA PROVENANCE + METRICS + CLINICAL OUTCOME LAYER
MLOps/Federated loop -> DRIFT -> RETRAIN -> CALIBRATE -> FAIRNESS -> APPROVE -> DEPLOY
```

4. Functional Requirements

ID	Requirement	Statement
FR-01	Input ingestion	Accept text, image, PDF and multimodal requests and create a common RequestContext.
FR-02	Input validation	Validate type, integrity, security, image quality, page extraction, OCR and content safety.
FR-03	Intent/relevance routing	Determine user intent, modality, cancer relevance, category, supported capability and phase status.
FR-04	ML gate	Permit ML inference only when modality/category/model/quality/phase constraints are satisfied.
FR-05	ML inference	Return prediction, model/version, class probabilities where supported, confidence, uncertainty and artifacts.
FR-06	Calibration	Maintain category/modality-specific calibration artifacts and versioned thresholds.
FR-07	OOD detection	Detect inputs outside expected data distribution and trigger review or abstention policies.
FR-08	Safe abstention	Return ABSTAIN/REVIEW or NO_GROUNDED_ANSWER when evidence or model certainty is insufficient.
FR-09	RAG retrieval	Support metadata filtering, hybrid retrieval, reranking, evidence scoring and citation identity.

ID	Requirement	Statement
FR-10	Local LLM	Structure and summarize supplied evidence; it shall not fabricate missing clinical evidence.
FR-11	External fallback	Use approved sources/APIs only when internal evidence is insufficient and external lookup is permitted.
FR-12	Answer verification	Verify relevance, grounding, provenance, contradiction, safety and citation integrity.
FR-13	Clinical XAI	Produce concept-based explanations linked to clinical findings and supporting/counter evidence.
FR-14	Technical XAI	Provide SHAP/Grad-CAM/attribution artifacts when supported by the model type.
FR-15	Explanation evaluation	Store fidelity, robustness, localization and clinician-utility evaluation results.
FR-16	Longitudinal timeline	Store time-stamped diagnosis, imaging, pathology, laboratory, treatment, response and outcome events.
FR-17	Temporal context	Use authorized longitudinal context in prediction/reasoning and expose which timepoints influenced output.
FR-18	Fairness evaluation	Compute subgroup performance, calibration, error and abstention metrics under approved governance.
FR-19	Fairness reporting	Produce machine-readable and human-readable fairness reports and flag unacceptable disparities.
FR-20	Human-in-loop	Allow clinician accept/edit/reject/request-evidence/abstain actions and record feedback.
FR-21	Clinical outcome capture	Capture study-specific workflow and patient-relevant outcome variables under approved governance.
FR-22	External validation	Support unseen-site, temporal and subgroup evaluation datasets without training leakage.
FR-23	Federated training	Support local site training and secure aggregation without raw patient-data movement.
FR-24	Continuous learning	Support staged retraining, validation, fairness, calibration and rollback gates.
FR-25	Drift monitoring	Detect data, performance, calibration and subgroup drift.
FR-26	Model registry	Store model ID, version, category, modalities, datasets, metrics, calibration, uncertainty, XAI and deployment state.

ID	Requirement	Statement
FR-27	Dataset registry	Store dataset version, population, taxonomy, modalities, preprocessing, annotation policy, subgroup metadata and splits.
FR-28	Case/report management	Create cases, attach inputs/results/evidence, generate reports and retrieve history by permission.
FR-29	Auditability	Retain route, model, evidence, source, explanation, uncertainty, fairness, clinician action and outcome provenance.
FR-30	Terminal safety states	Support explicit unsupported, abstain, review-required and no-grounded-answer states.

5. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Reliability	Critical services shall fail safely; uncertain or unsupported medical claims shall not be silently emitted.
NFR-02	Performance	Latency budgets shall be benchmark-derived for ingestion, routing, inference, retrieval, XAI, LLM and external lookup.
NFR-03	Scalability	ML, RAG, longitudinal, outcome and reporting services shall scale independently.
NFR-04	Security	Use secure auth, RBAC, encryption in transit/at rest, secrets management, validation, rate limiting and vulnerability scanning.
NFR-05	Privacy	Minimize sensitive data, enforce access control, support de-identification and local data retention for federated workflows.
NFR-06	Explainability	Clinical explanations must be understandable, evidence-linked and accompanied by method/version metadata.
NFR-07	Calibration	Probability outputs used for decisions shall be evaluated for calibration and monitored after deployment.
NFR-08	Fairness	Supported models must have defined subgroup evaluation plans before clinical deployment.
NFR-09	Longitudinal integrity	Patient timelines must preserve immutability of historical facts, timestamps and provenance.

ID	Area	Requirement
NFR-10	Outcome integrity	Clinical outcome measurements must be linked to study protocol versions and avoid inappropriate leakage into model training.
NFR-11	Observability	Collect logs, metrics, traces, drift, calibration, fairness, outcome and audit signals without unnecessary PHI exposure.
NFR-12	Reproducibility	Dataset, model, policy, calibration, explanation and evaluation versions must be reproducible.
NFR-13	Interoperability	Support a standards-oriented integration path for hospital/EHR/PACS/RIS/LIS/FHIR/HL7 workflows where deployed.
NFR-14	Portability	Local, hybrid and cloud deployments shall be supported through containerized services and provider abstraction.

6. AI/ML Research Requirements

- All category heads must follow the authoritative taxonomy and documented task definition.
- Internal and external validation must be separated and leakage-controlled.
- Performance must be reported by overall cohort and approved subgroups.
- Calibration, uncertainty and OOD behavior must be reported alongside discrimination metrics.
- Model cards must include intended use, exclusions, limitations, data lineage, fairness results, uncertainty characteristics and explanation support.
- Continuous-learning updates must not enter production without repeat validation, fairness assessment, calibration check and rollback artifact.
- Research experiments must record random seeds, software versions, dataset versions, configuration, hyperparameters and evaluation scripts.

7. Clinical XAI Requirements

- Clinical explanations shall identify salient concepts, not only raw pixels/features.
- Each explanation shall expose source evidence and, when available, counter-evidence.
- Technical attribution maps shall be retained for audit but shall not be presented as inherently clinically valid.
- Explanation quality shall be evaluated with quantitative fidelity/robustness measures and clinician-centered usefulness studies.
- Explanation text shall distinguish model evidence from retrieved guideline/literature evidence and generated synthesis.

8. Longitudinal Requirements

Requirement	Specification
Timeline	Append-only time-stamped event model with diagnosis, imaging, pathology, labs, treatment, response, outcomes.
Temporal leakage control	Training/validation splits must respect time when the study question is prospective/forecasting.
Trajectory features	Support trends, changes, intervals, treatment exposure and recurrence/progression markers.

Requirement	Specification
Context visibility	UI/report must disclose which historical events were used by the model or reasoning layer.
Missingness	Missing events must be represented explicitly rather than treated as absence.

9. Fairness Requirements

- Define evaluation strata before study execution.
- Report subgroup sensitivity, specificity, F1/AUC, calibration error, false-negative rate, false-positive rate, abstention rate and explanation-utility results where sample size permits.
- Investigate whether model routing or data-quality gates disproportionately reject specific groups.
- Fairness reports must be versioned and attached to the model release record.
- Mitigation actions may include data collection, reweighting, subgroup-specific calibration, threshold adjustment or model redesign; mitigation must be justified and revalidated.

10. Clinical Outcome Requirements

Outcome level	Required measurements
Technical	AUC, sensitivity, specificity, F1, calibration, segmentation metrics as applicable, OOD/abstention performance
Workflow	time-to-diagnosis, time-to-review, clinician workload, decision time, override rate
Decision quality	appropriate escalation, unsafe misses, clinician-AI agreement, evidence use
Patient-level	study-specific downstream care/outcome variables approved by ethics/governance
Safety	critical-error rate, grounding failures, incorrect citation rate, unsafe automation rate

11. Data Model Additions

Object	New/updated fields
Prediction	calibration_set_version, confidence, uncertainty_vector, ood_score, abstention_decision
Explanation	explanation_id, concept_list, evidence_ids, counter_evidence_ids, method, fidelity_metrics
PatientTimelineEvent	event_time, event_type, source, provenance, confidence, missingness_state
FairnessEvaluation	model_version, subgroup_definition, metric_set, results, threshold_policy, status
ClinicalOutcome	study_id, endpoint, measurement_time, value, source, adjudication_state
ModelVersion	calibration_policy, uncertainty_policy, fairness_report_id, explanation_policy, drift_baseline

12. API Requirements

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POST /api/v2/analyze
POST /api/v2/ingest
POST /api/v2/predictions/{id}/review
GET  /api/v2/predictions/{id}/uncertainty
GET  /api/v2/predictions/{id}/explanation
GET  /api/v2/cases/{id}/timeline
POST /api/v2/cases/{id}/timeline-events
GET  /api/v2/models/{id}/fairness
GET  /api/v2/models/{id}/calibration
GET  /api/v2/models/{id}/drift
POST /api/v2/outcomes
GET  /api/v2/studies/{id}/outcomes
POST /api/v2/federated/rounds
GET  /api/v2/audit
```

13. Testing and Acceptance

- Unit, integration, API, security, routing, model smoke, RAG, XAI, longitudinal, fairness, calibration and drift tests.
- Synthetic and retrospective tests must include known ambiguous/unsupported inputs and required safe-failure states.
- External validation datasets must be held out from all model selection and threshold tuning.
- Clinical validation must have a predefined analysis plan and endpoint definitions.
- Phase promotion requires all relevant technical, uncertainty, fairness, safety and documentation gates to pass.

14. Phase Plan v2.0

Phase	Scope	Exit gate
Phase 1	Breast + Kidney; multimodal ingestion, routing, ML, uncertainty skeleton	Validated end-to-end pipeline + calibrated baseline
Phase 2	All eight categories; authoritative taxonomy and category heads	All categories routable + external validation protocol operational
Phase 3	RAG, KG, local LLM, concept-XAI, evidence verifier, longitudinal timeline	Grounded workflow + explainability evaluation + timeline integrity
Phase 4	Federated learning, fairness engine, drift monitoring, continual-learning controls	Multi-site simulation + fairness/uncertainty gates + rollback
Phase 5	Clinical pilot, outcome capture, comparative evaluation, hospital integration	Prospective evidence package + production-oriented baseline

15. Safety and Regulatory Position

OncoSentinel is an AI-assisted research and clinical decision-support platform. It must clearly distinguish model predictions, retrieved evidence, generated synthesis, and external information. It shall provide safe abstention and shall not autonomously prescribe treatment. Production clinical use requires institution-specific governance, validation, legal review, and regulatory assessment appropriate to the deployment jurisdiction and intended use.

16. Literature References

- Kourou, K., Exarchos, T. P., Exarchos, K. P., Karamouzis, M. V., & Fotiadis, D. I. (2015). Machine learning applications in cancer prognosis and prediction. Computational and Structural Biotechnology Journal, 13, 8-17.
- Pati, S., et al. (2022). Federated Learning Enables Big Data for Rare Cancer Boundary Detection. arXiv:2204.10836v2.

- Zhang, B., Shi, H., & Wang, H. (2023). Machine Learning and AI in Cancer Prognosis, Prediction, and Treatment Selection: A Critical Approach. Journal of Multidisciplinary Healthcare, 16, 1779-1791.
- Sangwan, H. (2024). Quantifying Explainable AI Methods in Medical Diagnosis: A Study in Skin Cancer. medRxiv preprint 2024.12.08.24318158.
- Shyr, C., Hu, Y., Tinker, R. J., et al. (2026). An artificial intelligence framework for end-to-end rare disease phenotyping from clinical notes using large language models. arXiv:2602.20324v1.